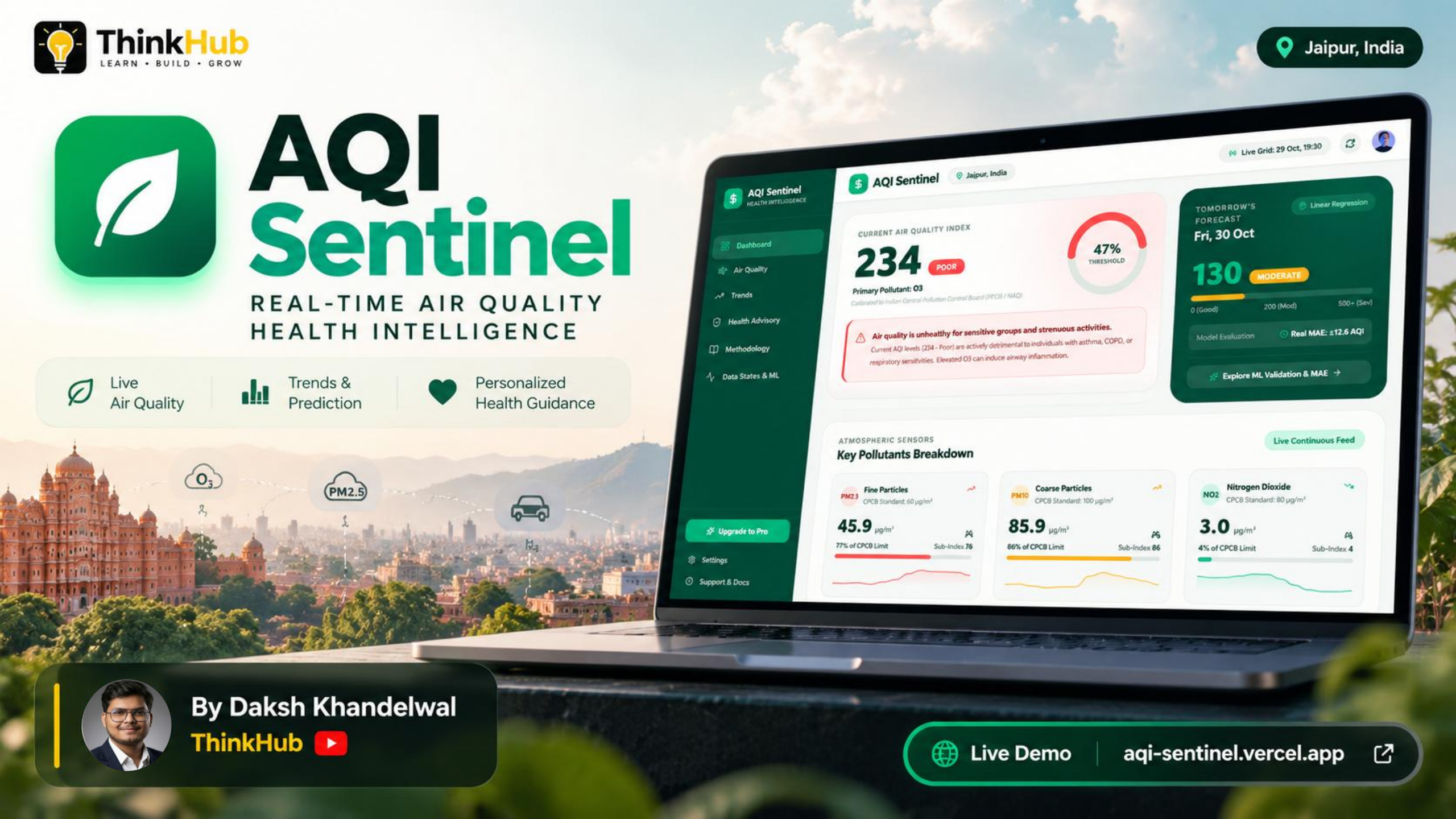
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Jaipur, India

AQI Sentinel

REAL-TIME AIR QUALITY HEALTH INTELLIGENCE

- Live Air Quality
- Trends & Prediction
- Personalized Health Guidance



By Daksh Khandelwal
ThinkHub

Live Demo | aqi-sentinel.vercel.app

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THE PROBLEM

Same Air, Different Impact

Air quality keeps changing, but raw numbers don't tell you what it means for you.



Air Quality Changes Throughout the Day



AQI can vary **significantly within a single day**

Raw Data is Confusing

PM2.5	45.9	µg/m³
PM10	85.9	µg/m³
O_3	182.0	µg/m³

?

- Is this good or bad?
- Which pollutant is the main concern?
- What should I do right now?

Different People, Different Risks



Children
More vulnerable



People with Asthma / COPD
Higher risk



Elderly People
More sensitive



Outdoor Workers
Higher exposure



Healthy Adults
Need clear guidance

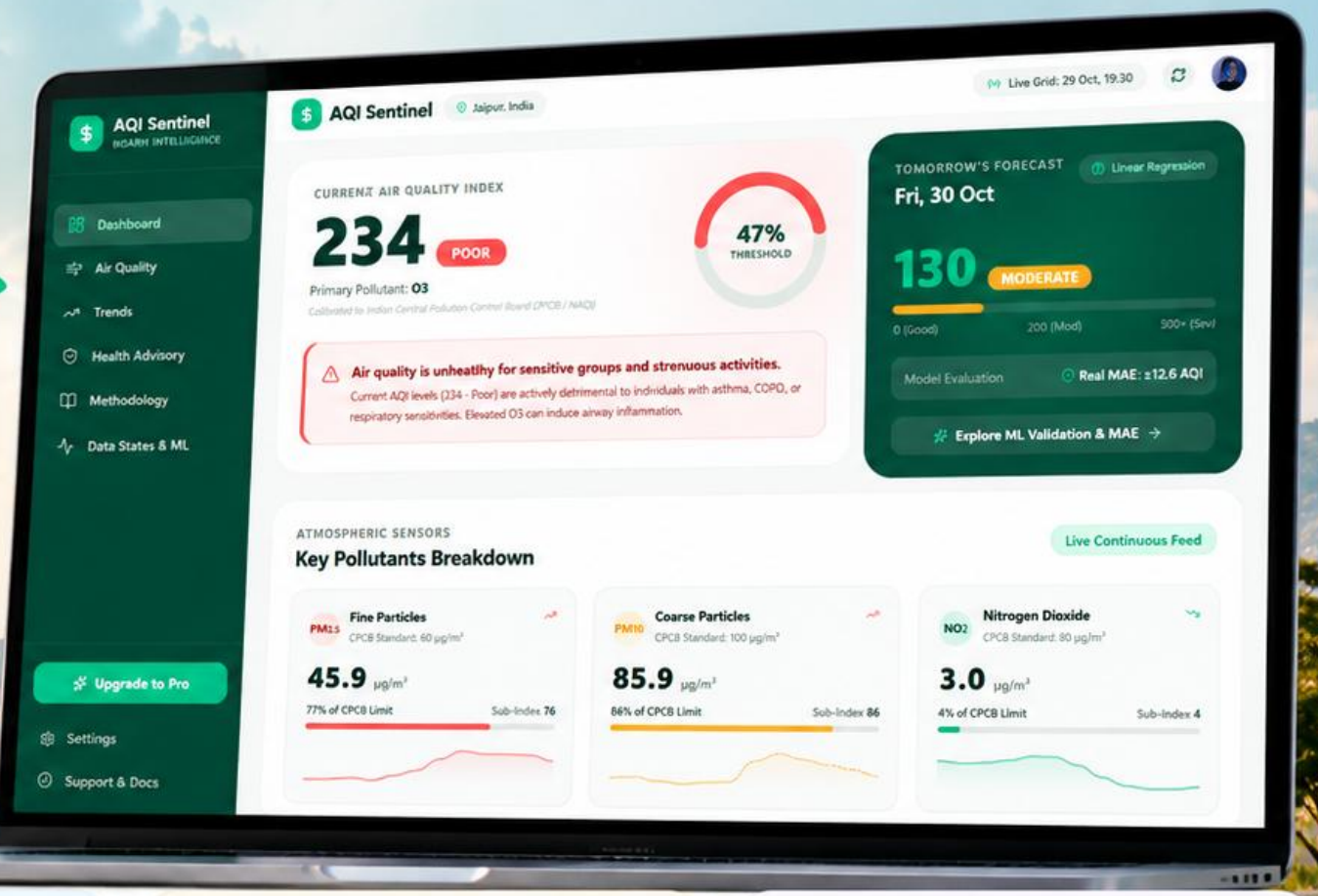
We need clear, understandable insights and practical precautions, not just raw pollution numbers.

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THE SOLUTION

From Data to Actionable Insights

AQI Sentinel turns complex air quality data into clear insights, predictions, and personalized guidance for your health.



1 Real-Time AQI & Pollutant Breakdown

Live air quality index with detailed PM2.5, PM10, NO2, O3, SO2 and CO measurements.

PM2.5 PM10 NO_2 O_3 SO_2 CO

2 Historical Trends & Analysis

Visualize how air quality changes over time with 24-hour, 7-day and 30-day views.



3 Next-Day AQI Prediction

Machine learning based forecast using temporal patterns and environmental data.



4 Profile-Based Health Guidance

Personalized recommendations for children, sensitive individuals, elderly people and outdoor workers.



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HOW IT WORKS

Jaipur, India

From Raw Data to Real Guidance

An end-to-end pipeline that converts real-time air-quality data into actionable, personalized health guidance for Jaipur.

1 Air-Quality Data

OpenAQ

Fetches real-time environmental telemetry for Jaipur (PM2.5, PM10, NO2, O3, SO2, CO).

2 Data Processing

Cleans and filters data, handles missing values and outliers, and prepares time-series features (lag and rolling averages).

3 AQI Analysis

AQI

Calculates pollutant sub-indices using CPCB / NAQI standards and determines overall AQI category (Good to Severe).

4 Next-Day Prediction

Uses Linear / Ridget Regression with lag-1, lag-2, lag-3 and rolling averages to predict tomorrow's AQI.

5 Personalized Advisory

Generates profile-based health guidance with practical precautions for different user groups.

Data Source

OpenAQ API

Global air-quality data platform

- Real-time environmental data
- Multiple pollutants
- Jaipur regional monitoring node



ML Model

Linear / Ridge Regression

Interpretable time-series model

Lag-1 | Lag-2 | Lag-3 | Rolling Avg

- Predicts next-day AQI
- Calibrated on Jaipur telemetry data
- Real MAE shown in the application

AQI Standards

CPCB / NAQI

Indian air-quality guidelines

- Good
- Satisfactory
- Moderate
- Poor
- Very Poor
- Severe

Profile-Based Guidance

Different users, different precautions

Children | Sensitive / Asthma | Elderly

Outdoor Workers | Healthy Adults

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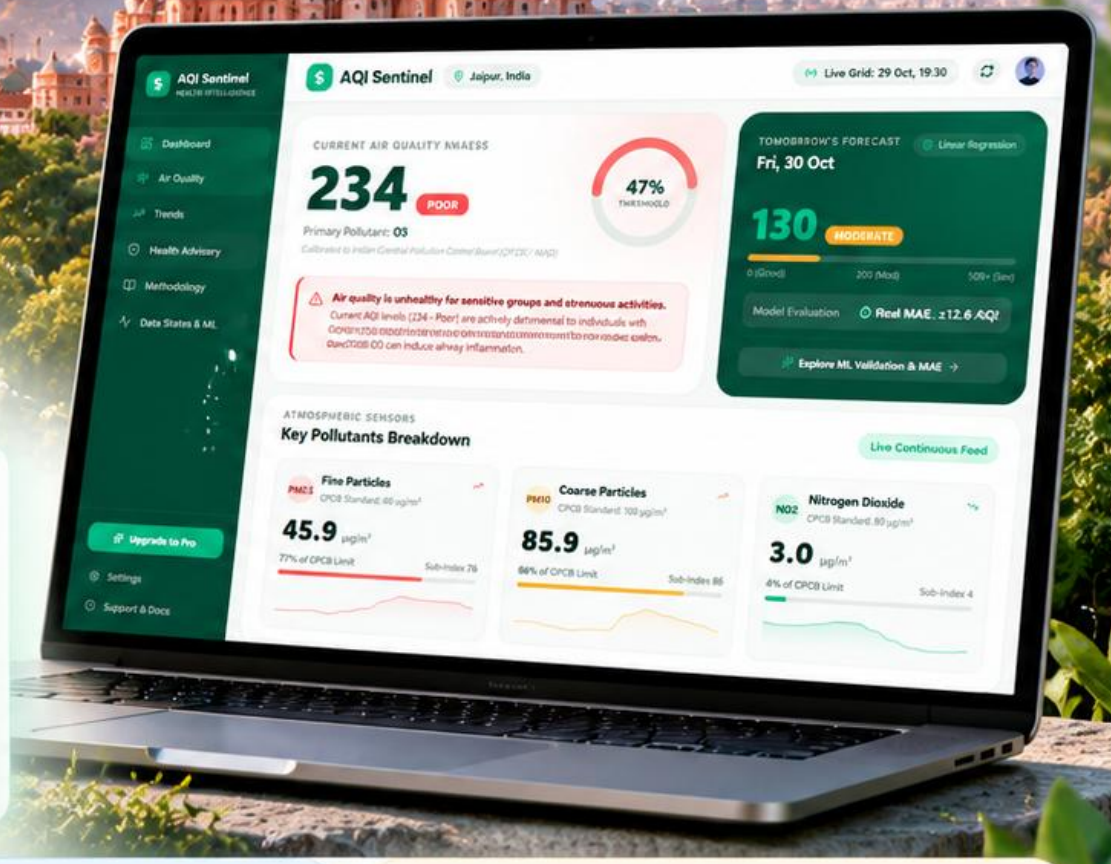
Jaipur, India

CONCLUSION

AQI Sentinel

REAL-TIME AIR QUALITY HEALTH INTELLIGENCE

A complete end-to-end web application that turns complex air quality data into actionable, personalized guidance for Jaipur.



Live AQI & Pollutant Data

Real-time air quality with PM2.5, PM10, NO2, O3, SO2 & CO.

Historical Trends

24-hour, 7-day and 30-day air quality analysis.

Next-Day Prediction

Linear Regression based AQI forecast with real MAE.

Personalized Health Guidance

Actionable advice for children, sensitive individuals, elderly and more.

Key Learning

How to take real-world data, apply data science and build a useful, visually compelling product with end-to-end deployment.

Project Links

Live Demo
aqi-sentinel.vercel.app

GitHub Repository
dk-khandelwal06/aqi-sentinel

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